PosiBuoy

Protecting and monitoring Posidonia, before it's too late

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Executive Summary

Posidonia oceanica meadows are among the Mediterranean's most valuable marine ecosystems, providing essential ecosystem services including **carbon sequestration**, **biodiversity support**, **shoreline protection**, and **water quality improvement**. Despite their ecological importance, these meadows are **under increasing pressure** from multiple human-induced and environmental stressors, including coastal infrastructure development, water pollution, recreational boat anchoring, rising sea temperatures associated with climate change, and excessive fishing activities.[1,2] Together, these pressures contribute to the **degradation and fragmentation** of seagrass habitats, threatening the long-term health of Mediterranean coastal ecosystems.

This project proposes a **smart eco-mooring system** that combines environmentally friendly mooring infrastructure with real-time environmental monitoring and digital management. Instead of allowing vessels to deploy traditional anchors, boats attach to permanently installed eco-mooring **buoys** secured using low-impact helical moorings,[3,4] eliminating one of the major direct causes of physical damage to Posidonia meadows. At the same time, each buoy serves as an autonomous environmental **monitoring station**, continuously measuring water quality and environmental conditions that may indicate pollution or ecosystem stress.

The system integrates underwater **imaging**, water-quality **sensors**, renewable **energy harvesting**, **edge Artificial Intelligence** (AI), and **cloud-based analytics** into a single autonomous platform. Edge AI processes underwater images locally to detect illegal anchoring and estimate Posidonia coverage, reducing communication bandwidth while enabling real-time monitoring. Environmental data are transmitted via the cellular network to a cloud platform, where more complex **predictive** AI models analyze long-term trends and identify early warning signs of meadow degradation. This enables marine park managers to implement **proactive conservation measures** before irreversible ecological damage occurs.

By simultaneously preventing anchor damage and monitoring other environmental stressors, such as declining water quality and climate-related changes, the proposed system supports more effective ecosystem management. The integrated reservation platform improves and **encourages** eco-mooring operations, while the continuous monitoring infrastructure generates verifiable **environmental data** that can support **scientific research**, evidence-based decision-making, and **blue-carbon certification** schemes. Overall, the proposed solution offers a scalable and financially sustainable approach to protecting Posidonia oceanica meadows and improving the management of Mediterranean marine protected areas.

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1 Introduction

Posidonia oceanica is an **endemic** seagrass species found throughout the **Mediterranean** Sea and is widely recognised as one of the region's most important marine ecosystems.[5,6] Its meadows provide numerous ecological services, including **carbon sequestration**, **oxygen production**, shoreline stabilisation, water quality improvement, and habitat for a wide range of marine species. As a result, the conservation of Posidonia meadows is essential for maintaining the ecological health and biodiversity of Mediterranean coastal waters.

Recent advances in renewable energy systems, and Artificial Intelligence (AI) have created new opportunities for continuous environmental monitoring and smarter management of marine protected areas. By integrating these technologies with sustainable mooring infrastructure, it is possible to both reduce human impacts on sensitive ecosystems and provide authorities with real-time environmental information to support evidence-based decision making.

This report presents the design of a **smart** eco-mooring **buoy system** that combines environmentally friendly mooring infrastructure with autonomous environmental monitoring, edge AI, cloud analytics, and a digital reservation platform. The proposed solution aims to improve the protection and long-term management of Posidonia oceanica meadows.

2 Problem Statement

Across the Mediterranean, particularly during the summer months and in areas surrounding ports and popular tourist destinations, Posidonia oceanica meadows are experiencing significant decline due to a combination of recreational boat **anchoring**, coastal **development**, water **pollution**, excessive **fishing**, and rising sea **temperatures** associated with climate change.[1,2] These pressures damage or degrade the seagrass ecosystem, reducing its ability to provide essential ecological services.

Posidonia meadows are among the Mediterranean's most valuable natural carbon sinks, storing approximately **fifteen times more carbon** and producing around **twice as much oxygen** per unit area as tropical rainforests.[7] They also provide critical **habitat** and nursery grounds for numerous fish, crustacean, and invertebrate species. Their continued decline therefore has significant environmental, ecological, and economic consequences.

Although several conservation measures have been introduced, including anchoring restrictions, eco-mooring installations, and seagrass restoration projects,[8,9] these initiatives have **not been sufficient** to halt the loss of Posidonia meadows. Furthermore, existing environmental monitoring is typically based on **periodic** field surveys rather than continuous observation, limiting the ability of marine authorities to detect ecosystem degradation early and implement proactive management measures.

3 Personas

To support our problem-solving process, we created **four personas**: Posidonia, a Seahorse, a Weekend Skipper, and a Coastal Ecosystem Guardian. These personas represent the key stakeholders affected by the problem, including both marine life and human users.

The goal was to ensure that **all relevant perspectives** and needs were considered throughout the design process, allowing us to develop a solution that balances environmental protection with human needs and is acceptable to everyone involved.

3.1 Posidonia

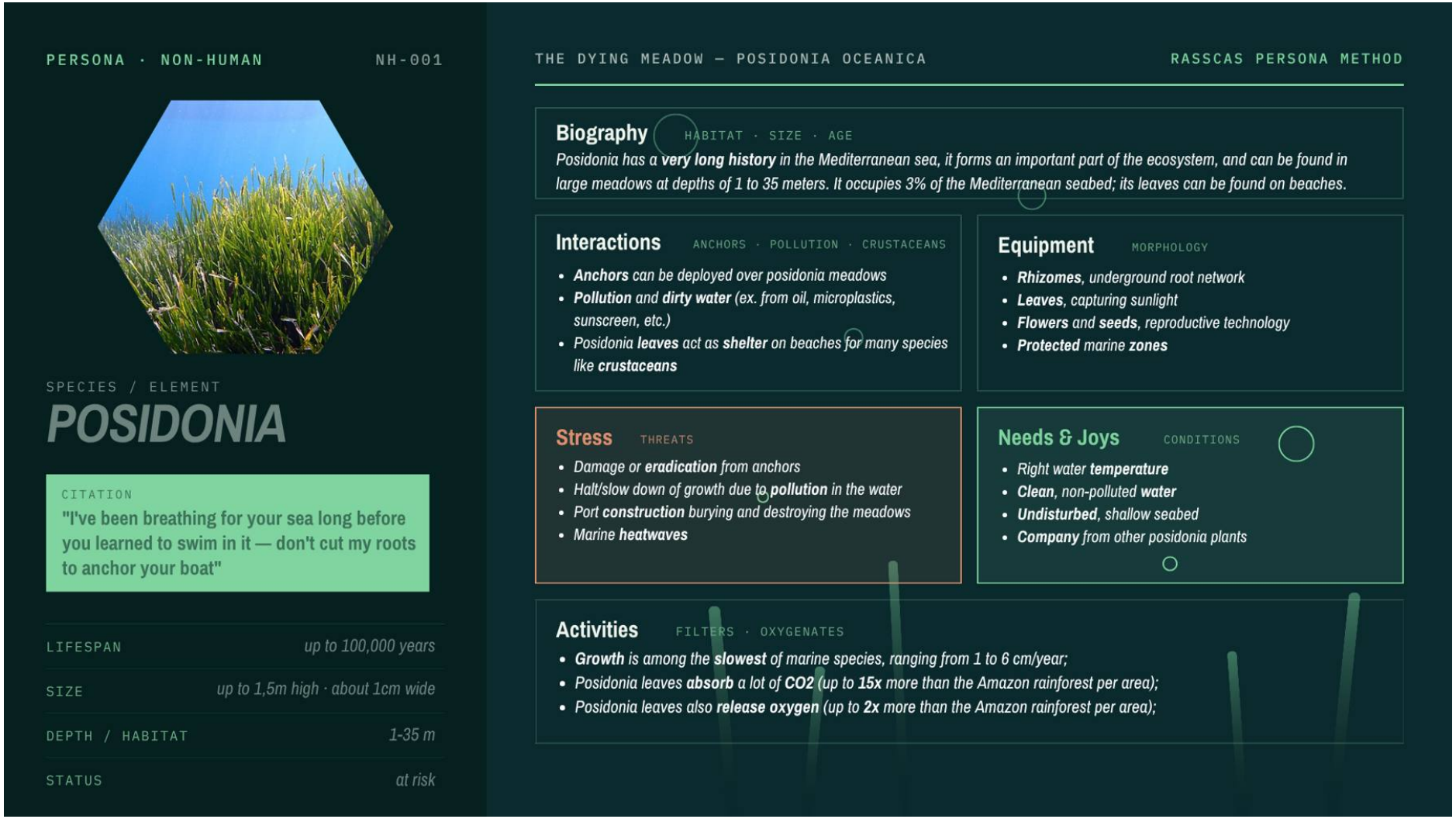


Figure 1: Persona: *Posidonia oceanica*

3.2 Seahorse

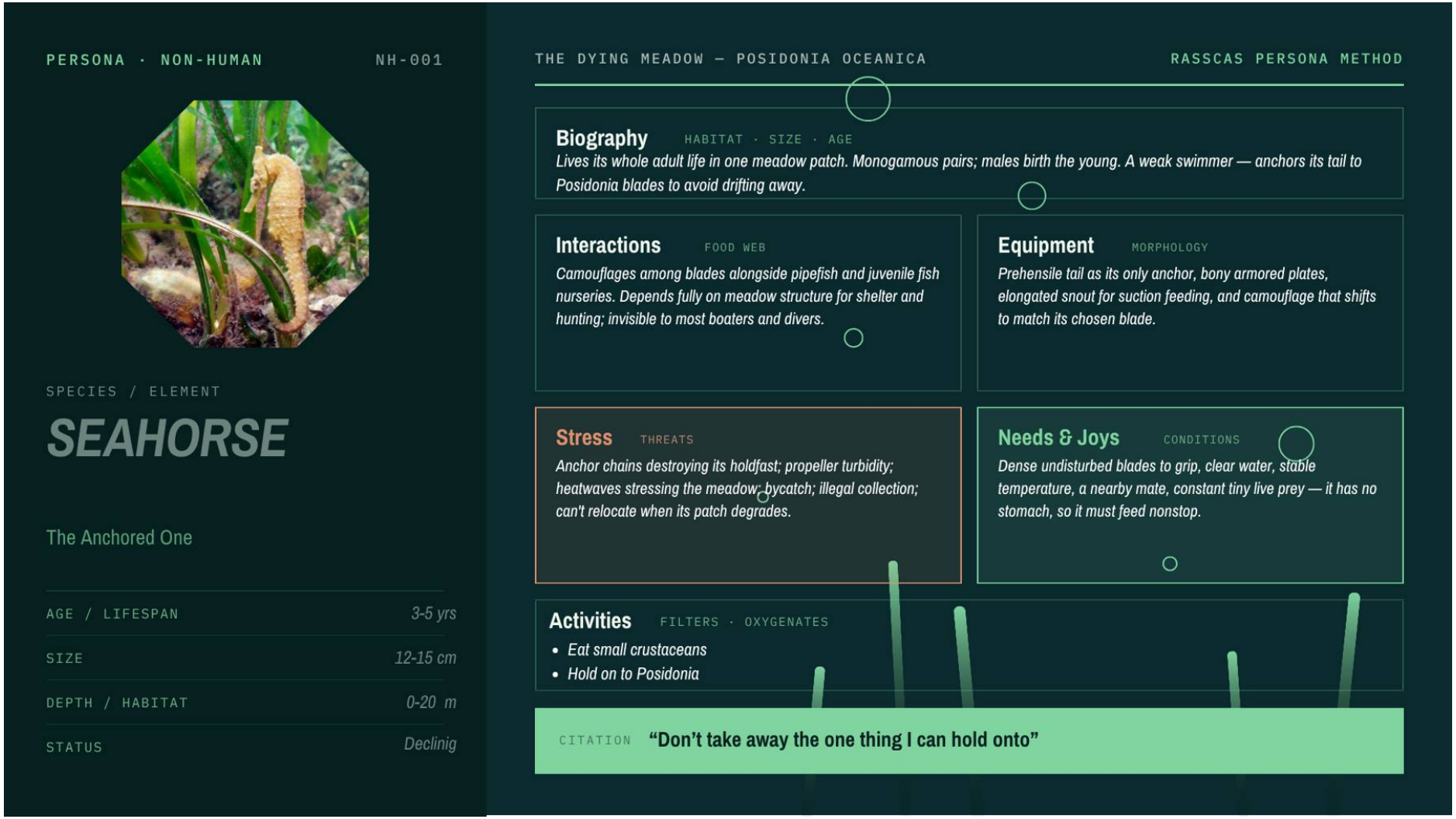


Figure 2: Persona: the seahorse

3.3 Weekend Skipper

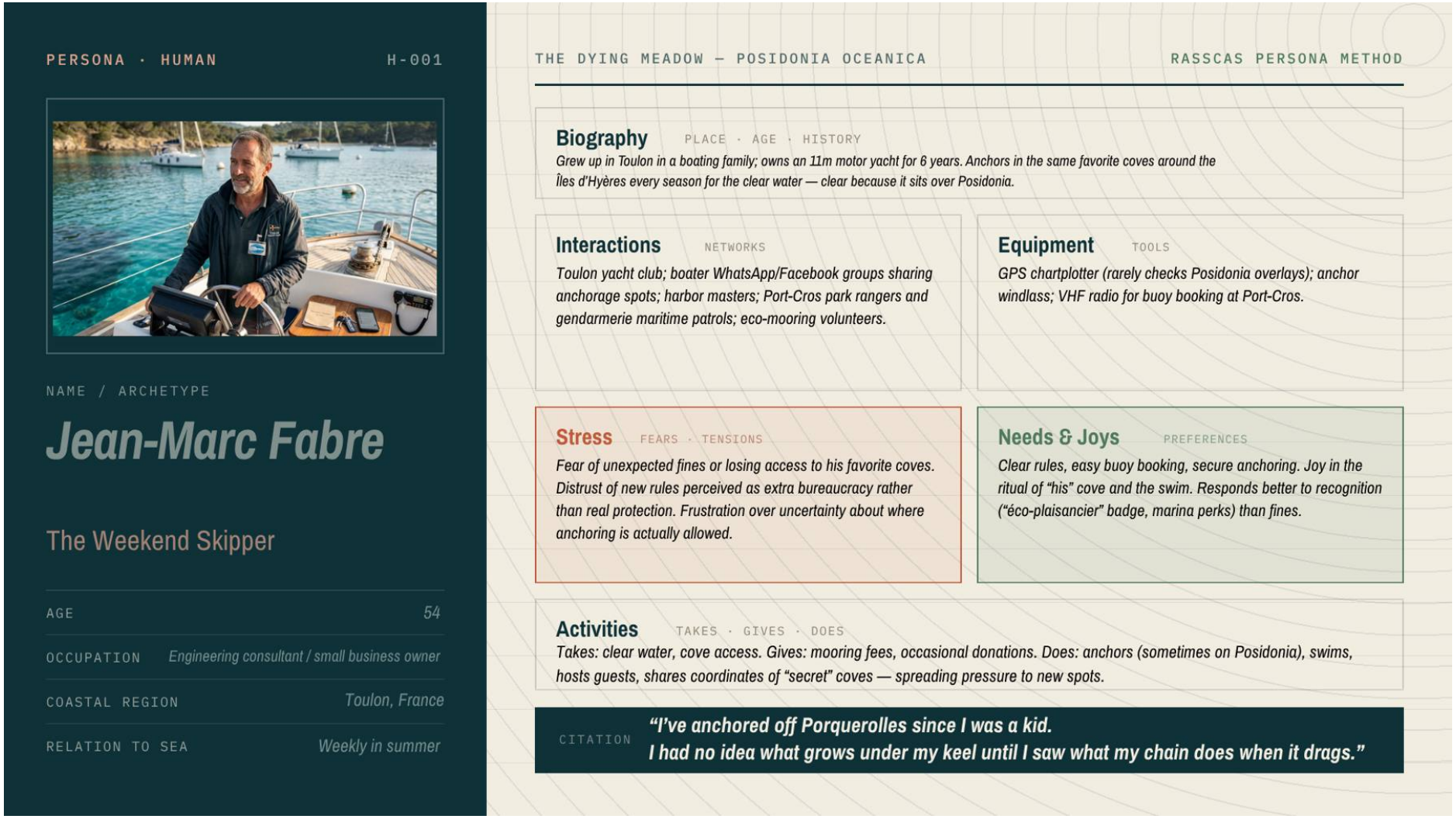


Figure 3: Persona: Jean-Marc Fabre, the Weekend Skipper

3.4 Ecosystem Guardian

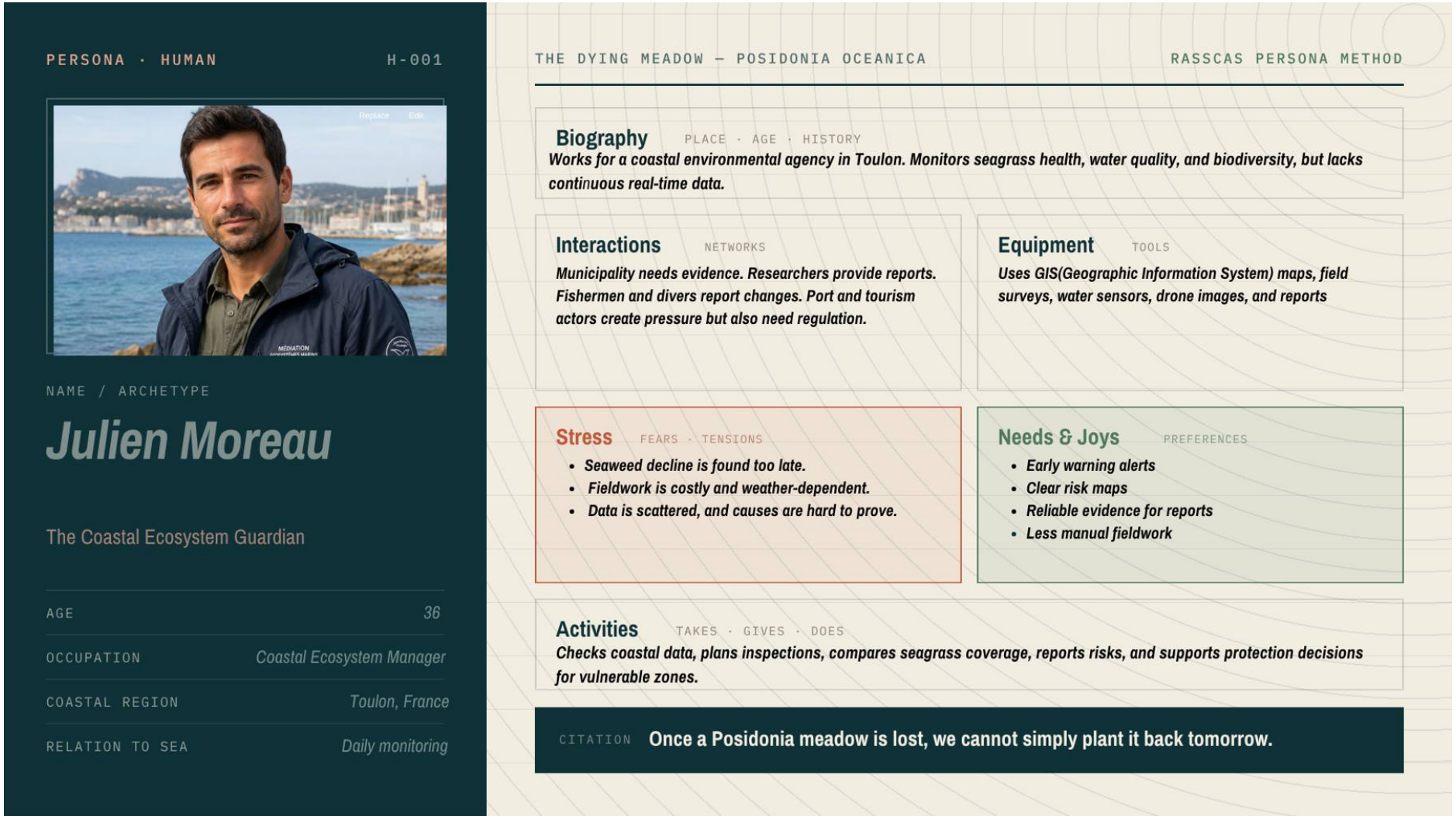


Figure 4: Persona: Julien Moreau, the Coastal Ecosystem Guardian

4 User Journey

To identify the features our product should provide, we created user journeys for each persona, illustrating their experiences over one day or one month **without** the PosiBuoy. We then developed corresponding user journeys showing how these experiences would change **with our solution** in place. The figures below present selected excerpts from these user journeys.

4.1 Current Scenario

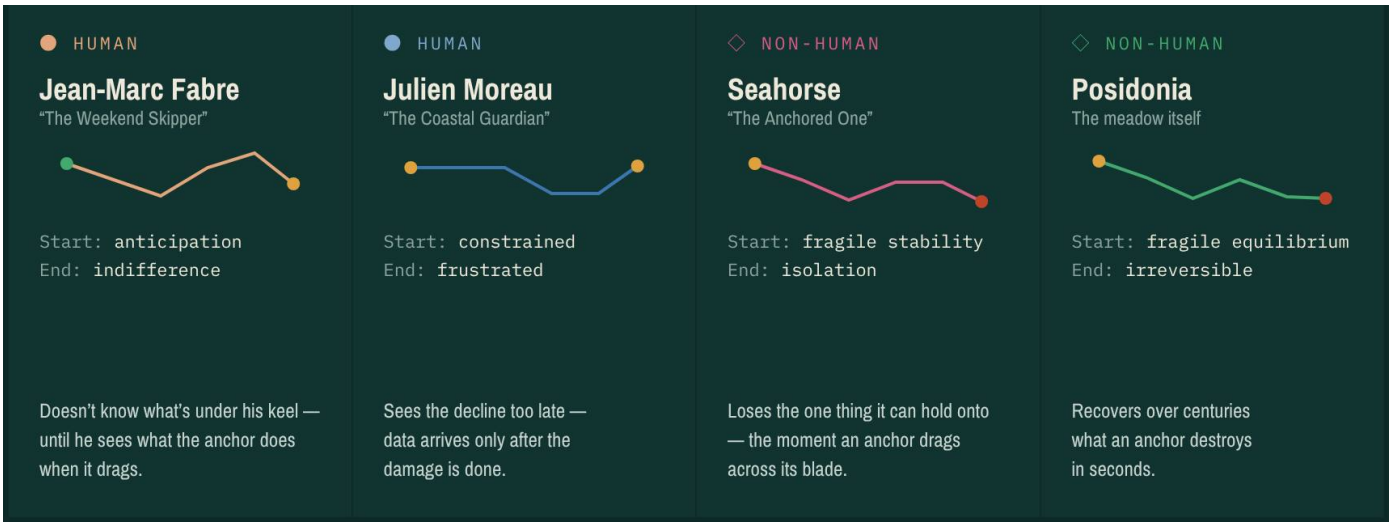


Figure 5: User journey before using the PosiBuoy

4.2 Proposed Scenario

Time	Actor	Action	Mood	Tool
08:00	Boat Owner 3	Reserves a buoy for a time slot.	5	Smartphone
10:00–10:20	Boat Owner 1	Attempts to anchor, is guided to an available buoy, and moors safely.	4	Smartphone
15:00–15:25	Guardian	Monitors occupancy and environmental data, detects pollution, and initiates a response.	3	PC
16:00–16:15	Boat Owner 2	Finds no available buoy and anchors illegally.	1	Smartphone
16:21	Guardian	Receives an alert for illegal anchoring, informs the Coast Guard, and moves away.	2	PC
16:42	Boat Owner 2	Receives a fine from the Coast Guard.	1	Coast Guard
18:30	Guardian	Confirms that pollution levels have returned to normal.	5	PC
18:43	Boat Owner 3	Navigates to the pre-booked buoy and moors.	5	Smartphone
19:00	Boat Owner 1	Leaves the buoy, making it available again.	5	–

Table 1: User journey after using the PosiBuoy

5 Solution

This chapter presents the proposed smart eco-mooring system from three complementary perspectives: the overall solution concept, the business model, and the technical implementation.

It describes how the system addresses the degradation of Posidonia oceanica meadows, how it can be economically sustained, and how the required technologies enable its operation.

5.1 Proposed Solution

The proposed solution is a smart eco-mooring system designed to protect *Posidonia oceanica* meadows while improving the management of recreational boating in marine protected areas. The system replaces conventional anchoring with fixed eco-mooring buoys, eliminating the need for vessels to deploy anchors that can damage the seabed.

Boaters reserve an available buoy through a mobile application, which displays real-time occupancy and allows users to pay the required mooring fee. Upon arrival, vessels attach directly to the buoy rather than dropping an anchor. This approach significantly reduces physical disturbance to the seagrass ecosystem while providing a more organized mooring infrastructure.

Each buoy continuously monitors environmental conditions using integrated sensors and underwater imaging. The collected data provide marine park authorities with real-time information about water quality and the health of surrounding *Posidonia* meadows. Edge Artificial Intelligence (AI) performs local image analysis to detect illegal anchoring and estimate seagrass coverage, minimizing the amount of data transmitted over the cellular network.

Historical environmental data are analysed in the cloud using predictive AI models capable of identifying early signs of meadow degradation. These predictions enable authorities to implement preventive management measures, such as temporarily closing vulnerable areas or redirecting vessel traffic before irreversible damage occurs.

The system is supported by a mobile application, cloud platform, and monitoring dashboard, creating an integrated solution for environmental protection, digital mooring management, and long-term ecosystem monitoring.

5.2 Technical Solution

This section describes the technical architecture of the proposed smart eco-mooring system. The architecture integrates environmental sensing, underwater imaging, renewable power generation, wireless communication, edge artificial intelligence (AI), and cloud-based analytics into a single autonomous platform. Together, these components enable continuous environmental monitoring, intelligent mooring management, and long-term assessment of *Posidonia oceanica* meadows while minimizing maintenance requirements and reducing the ecological impact of recreational boating.

The following subsections present the main technical components of the system, beginning with the environmental sensing hardware and continuing through the communication, power, and AI subsystems that support the overall functionality of the buoy network.

5.2.1 Environmental Monitoring

Each buoy incorporates a submerged environmental monitoring unit positioned below the waterline,[10,11] The unit continuously measures key water-quality parameters relevant to the health and status of the surrounding marine ecosystem.
The sensor suite can be configured according to site-specific monitoring objectives and stakeholder requirements. Different combinations of sensors may therefore be installed depending on the intended application. As maintenance requirements vary by sensor type, the selected configuration may affect the required maintenance interval.

Following is a list of possible sensors that can be included:

- Underwater camera
- Water temperature sensor
- pH sensor (requires frequent maintenance)
- Turbidity sensor
- Dissolved oxygen sensor[12]
- Salinity (electrical conductivity) sensor
- Nutrient pollution sensor (e.g., nitrate and phosphate concentrations)
- Load cell for boat occupancy detection

Together, the network provides a spatially distributed, real-time monitoring grid capable of assessing environmental conditions across the entire mooring field.

5.2.2 Seabed Imaging

Each buoy can be equipped with a downward-facing underwater camera that continuously monitors the surrounding seabed. Together with water quality sensors, the camera forms a sensor unit located below the water surface. This unit is connected to the buoy, which provides power and enables data transmission. Combined with edge AI, the camera system serves two primary purposes:

- Long-term assessment of Posidonia oceanica coverage and meadow health
- Illigal anchoring detection

To maintain image quality in the marine environment, the optical system incorporates a mechanical lens-wiper mechanism, as is also offered on commercially available marine sensor systems.[13] The wiper periodically removes algae and biofilm accumulation before sensor effectiveness is compromised.
Underwater cameras are also used on established marine monitoring platforms, such as the UK's Centre for Environment, Fisheries and Aquaculture Science (CEFAS) SmartBuoy network.[14,15,16]

As an additional passive anti-fouling measure, the optical ports are surrounded by a copper or copper-alloy housing. Copper naturally inhibits the growth of marine fouling organisms, reducing biological accumulation between cleaning cycles.

Despite these mitigation measures, routine maintenance remains necessary. Manual servicing is expected approximately every 1-4 months[17] (depending on algae seasonality) to remove any residual fouling and verify camera performance. This represents a predictable and manageable maintenance schedule rather than continuous intervention.

5.2.3 Digital Platform

Each buoy continuously reports its occupancy status to a central management platform. A companion mobile application enables boaters to view real-time buoy availability on an interactive map, reserve a mooring in advance, and pay the associated mooring fee. This operational model is comparable to existing paid eco-mooring systems, such as ZMEL (Zones de Mouillages et d'Équipements Légers),[18] already implemented within French marine protected areas.

The buoys can also be equipped with integrated loudspeakers to provide automated notifications, including reminders when reservation periods have expired and warnings to vessels detected

entering protected seagrass areas. Regulatory enforcement remains the responsibility of harbour police operating under municipal authority.

5.2.4 Power System

The buoy operates using a hybrid renewable energy system to maximise reliability throughout the year. Solar photovoltaic panels provide the primary energy source during summer months, when solar irradiance is highest and sea conditions are generally calmer.

To compensate for reduced solar generation during winter, each buoy incorporates a wave-energy harvesting system. An internal oscillating-mass generator converts the buoy's vertical motion into electrical energy,[19] taking advantage of increased wave activity during periods of rougher seas. These two energy sources are naturally complementary, with seasonal variations in wave energy offsetting reductions in solar production.

Electrical energy is stored within an onboard battery bank, ensuring uninterrupted operation during prolonged periods of low sunlight and limited wave activity.

5.2.5 Communication

Environmental measurements, occupancy information, and system diagnostics are transmitted via the 4G/5G cellular network or LoRaWAN (depending on site characteristics) to a central cloud-based platform. This enables real-time monitoring, remote system management, and integration with the reservation application.

5.2.6 Predictive Environmental Management

Data collected by each buoy—including water-quality measurements, camera-derived estimates of Posidonia coverage, hydrodynamic conditions, and vessel activity—are aggregated into a time-series database.

These datasets can be used to train predictive models capable of identifying early indicators of meadow degradation before visible ecological damage occurs. By detecting emerging trends, the system enables proactive management actions, such as temporarily closing vulnerable mooring areas or redirecting vessel traffic away from stressed sections of the meadow, thereby reducing the likelihood of long-term habitat degradation.

5.2.7 Maintenance

As the buoys are deployed year-round in a harsh marine environment, regular preventive maintenance is required to ensure the reliability of the sensing, imaging, communication, and power subsystems. In particular, submerged cameras and water-quality sensors are exposed to biofouling, sediment accumulation, corrosion, and gradual calibration drift.

The maintenance interval depends on the installed sensor configuration, as different instruments require different levels of cleaning, calibration, and component replacement. The intervals should then be also adapted to local environmental conditions, sensor manufacturer specifications, seasonal algae growth, and the required accuracy of the monitoring programme.

5.3 Funding and Revenue Model

The proposed business model combines one-time sales with recurring subscription-based revenue, creating both immediate income and long-term financial sustainability.

5.3.1 One-Time Revenue

- **Sale of Smart Eco-Mooring Buoys:** Revenue is generated through the sale of the buoy system to municipalities, marina operators, marine protected areas, and environmental agencies.
- **Installation Service (Optional):** Customers may purchase an installation service carried out by certified partner contractors, including deployment of the helical eco-mooring[3] and commissioning of the buoy.

5.3.2 Recurring Revenue

- **Maintenance Contract:** Annual subscription covering preventive maintenance, sensor calibration, cleaning, hardware inspections, and software updates.
- **Edge AI Analytics Subscription:** Optional subscription providing access to predictive environmental analytics, automated event detection, and AI-generated summaries of environmental conditions.
- **Data-as-a-Service (DaaS):** Subscription-based access to real-time and historical environmental datasets collected from buoy networks operated by the company, serving researchers, environmental agencies, and coastal managers.
- **Carbon Credit Verification Service:** Service fee charged for each carbon credit verification cycle. Environmental monitoring data generated by the buoy network supports verification and auditing of blue-carbon projects protecting *Posidonia oceanica* meadows.[20]
- **Boater Reservation Fees:** Boat owners pay a fee for each buoy reservation made through the mobile application. A refundable security deposit may also be required to discourage no-shows and ensure compliance with reservation terms.

5.4 Architecture Diagram

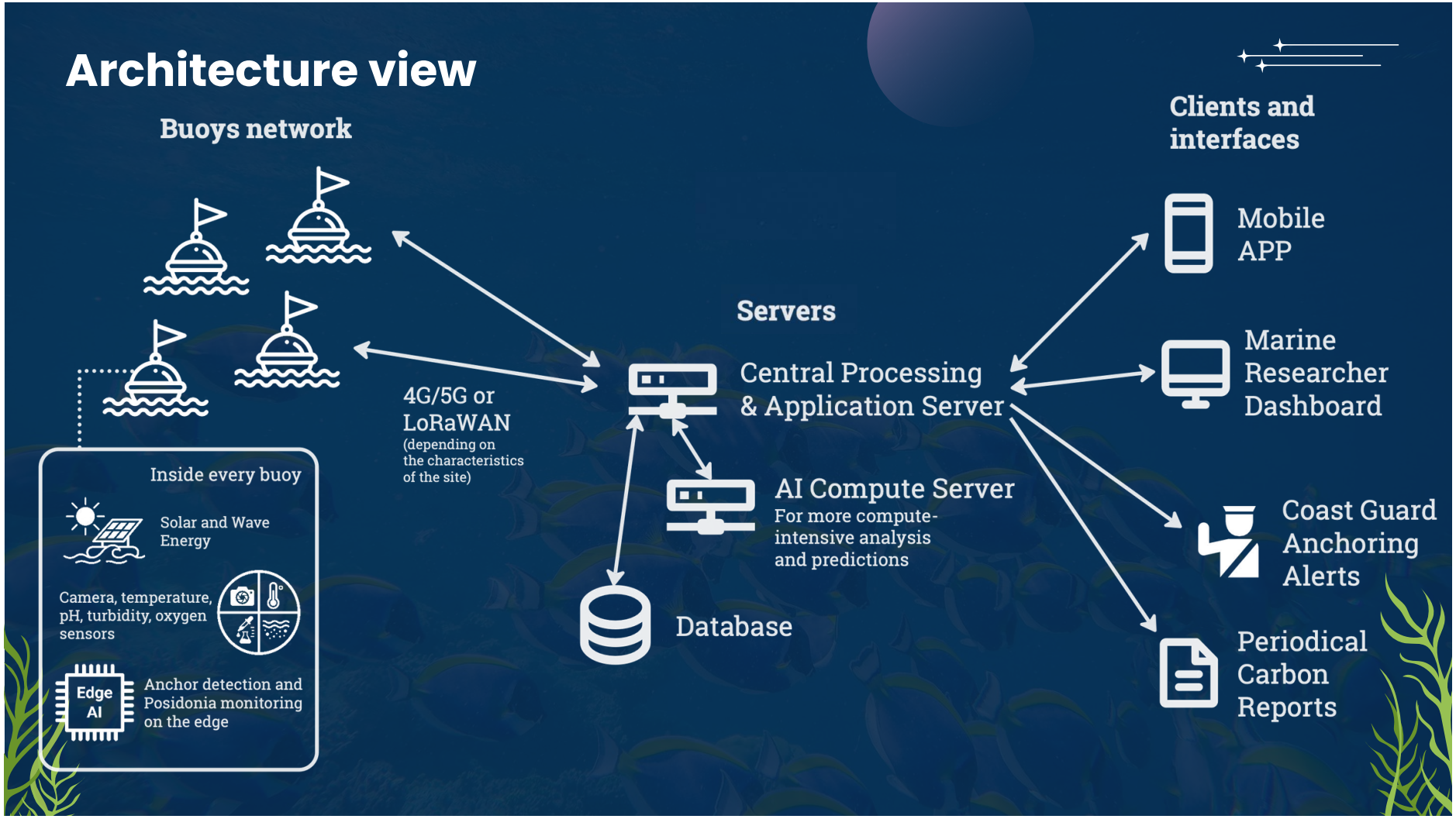


Figure 6: System architecture: buoy network, cloud services, and client interfaces

5.5 Buoy Architecture

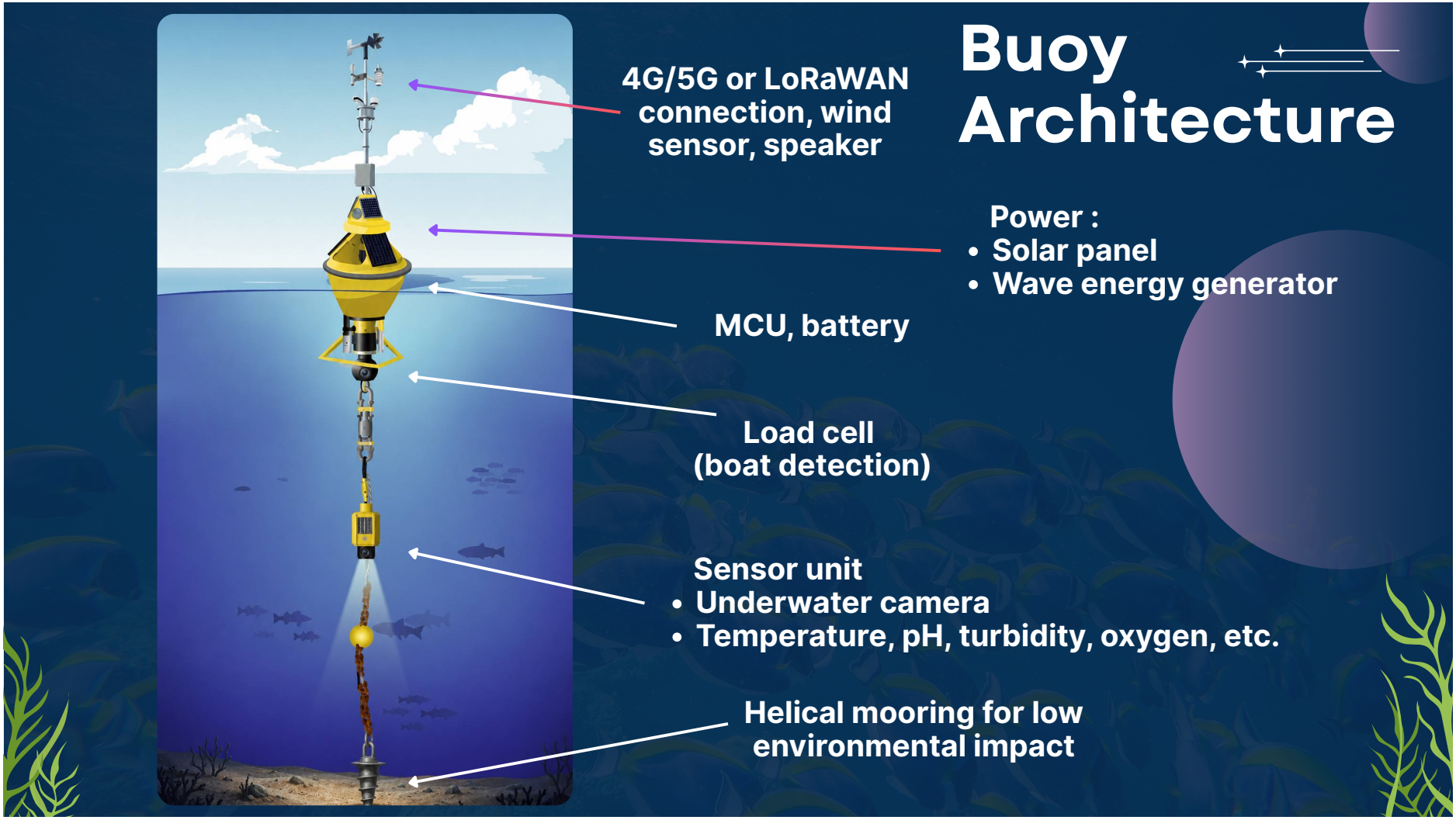


Figure 7: Buoy architecture and on-board subsystems

5.6 Business Model Canvas

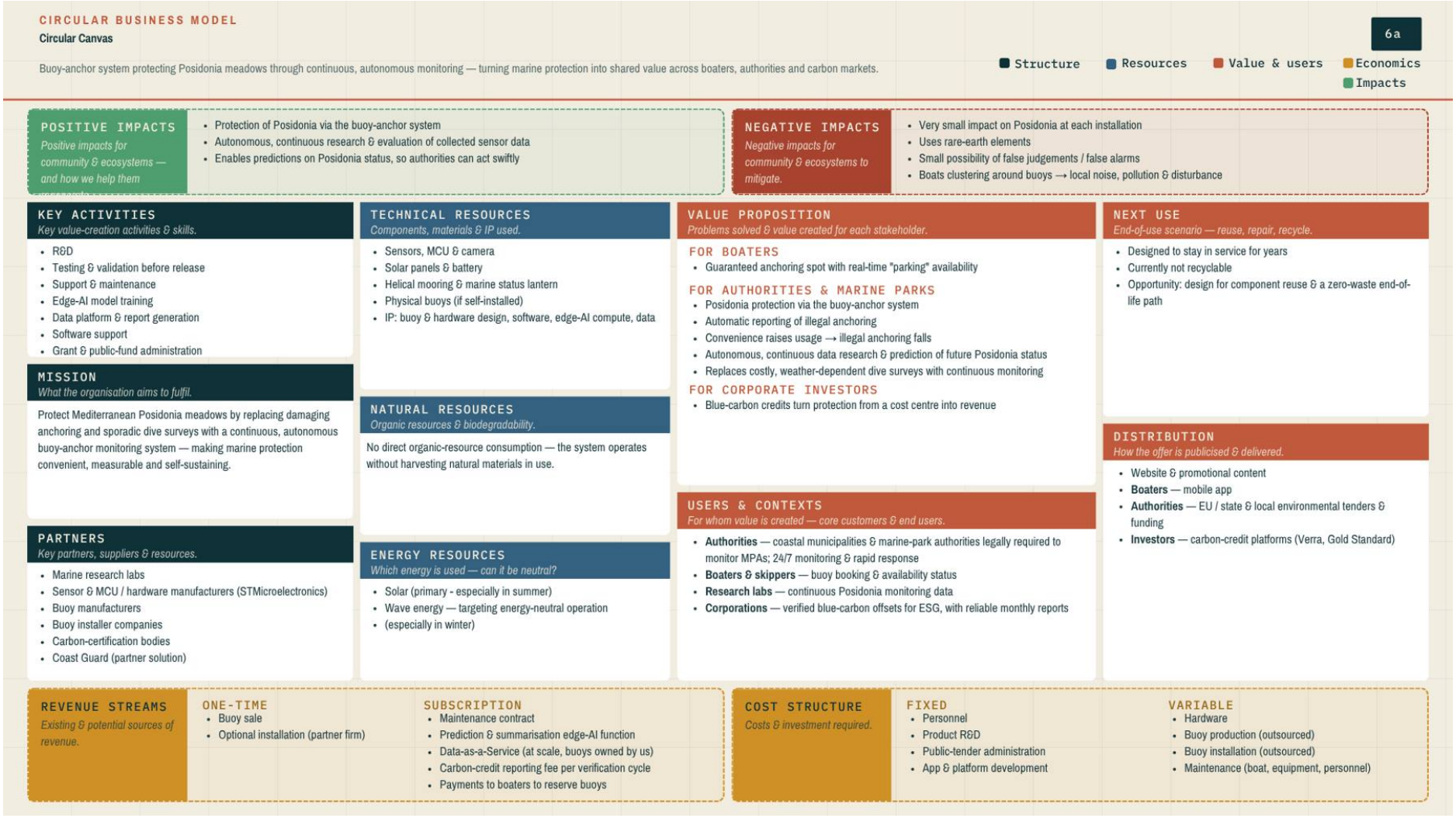


Figure 8: Business Model Canvas for the PosiBuoy system

6 Project Management

6.1 Work Management

What	Who	Why
Concept	everybody	Core project vision
Personas	everybody	Everyone has different perspectives
Architecture	Flavio, Paula	technical and system knowledge
Business Model Canvas	Flavio, Paula, Dongrui, Xiaoshan	Business and market analysis skills
User Journey	Flavio, Paula, Dongrui, Xiaoshan	Close involvement in user research
Design	Stefano, Andrea	design expertise
Proof of Concept (App)	Stefano, Andrea	Software development experience
Proof of Concept (Dashboard)	Flavio, Stefano, Andrea	implementation skills
Proof of Concept (Prototype)	Flavio	Embedded systems major
Presentation	everybody	
Hook	Dongrui	right tools and skills
Report	Paula, Xiaoshan	right skills

6.2 Problems

One of the main challenges during the project was communication within the team. Team members often worked independently on their respective tasks and made decisions that were necessary for their work. However, some of these decisions were not communicated clearly to the rest of the group, leading to inconsistencies between different project components. These issues were typically resolved through regular group discussions.

Another challenge was maintaining up-to-date project documentation. The shared notes and project files were not always updated after major decisions, making it difficult for all team members to stay aligned with the latest developments.

Finally, as English was not the native language of any team member, occasional misunderstandings occurred during discussions. Although these were generally minor, they sometimes required additional clarification to ensure a common understanding.

7 Results

For the proof of concept, a mobile application, a Guardian Dashboard, and a prototype of the PosiBuoy were developed. The mobile application demonstrates the reservation workflow for the smart eco-mooring system, while the dashboard provides an overview of environmental and operational data. The prototype showcases several of the sensing capabilities envisioned for the final PosiBuoy.

7.1 App

The developed web application demonstrates the proposed reservation and monitoring system for the smart eco-mooring buoys. Although the application is a simulation, it illustrates the intended user experience and the main operational features of the system.

Users reserve a time slot within a protected bay rather than selecting a specific buoy. The day is divided into three reservation periods: morning, afternoon, and night. Upon arrival, the system automatically assigns an available buoy within the selected bay. Allocation follows a first-come, first-served policy among users holding valid reservations.

To maximize buoy utilization, the reservation system incorporates an overbooking strategy based on an estimated no-show rate, similar to practices used in the airline industry. Overbooking is applied at the bay and time-slot level instead of to individual buoys. Reservations can be cancelled free of charge up to a configurable period before the start of the reservation. Late cancellations are treated as no-shows and are included in the overbooking model.

The check-in procedure combines automatic boat detection, a load cell, with manual confirmation in the application. If a user fails to check in within the allowed time window, the reservation is released and the buoy becomes available for reassignment within the same bay and time slot.

To discourage misuse of the system, each reservation requires a refundable security deposit. The deposit is forfeited in the event of a no-show or when violations are detected, such as mooring outside the assigned reservation period or outside the designated zone.

The application also integrates environmental data collected by the smart buoys. Boat owners can view real-time weather conditions and water quality measurements before and during their reservation. Wind measurements serve a dual purpose: they provide safety-related information and support a predictive occupancy model. Favorable wind forecasts are associated with increased boating activity and can therefore be used to adjust reservation prices dynamically. The dynamic pricing mechanism only affects the booking fee, while the security deposit remains unchanged.

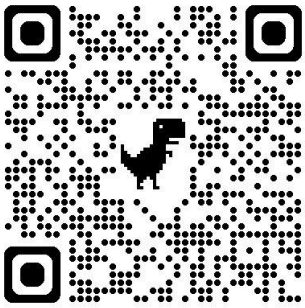


Figure 9: App QR-Code

Screenshots of the mobile application prototype are in Appendix A

7.2 Data Dashboard

The simulated Guardian Dashboard provides a comprehensive overview of the monitored protected area. Environmental sensor data can be viewed both at the bay (section) level and for individual smart buoys. In addition to simulated real-time water quality and wind measurements, the dashboard displays the current occupancy status of the buoys, predictions of the health status of the Posidonia meadow, and the operational status of each buoy.

To support decision-making, each environmental parameter is accompanied by an interpretation indicating whether the measured value is within the normal range or classified as slightly above, above, slightly below, or below the expected conditions for that location. This allows the Guardian to quickly identify anomalies and assess the environmental state of the protected area.

The Guardian Dashboard can be accessed through the following link:

`https://posi-buoy-guardian.vercel.app/`

Screenshots of the Guardian Dashboard prototype are in Appendix B.

7.3 Prototype

For the prototype, we used a Nucleo-L152RE board, with an STM32 microcontroller, together with several sensors and actuators, including a temperature sensor, distance sensor, turbidity sensor, wind speed sensor, and three LEDs.

For the proof of concept, the turbidity was measured using a simple setup consisting of a light source passing through the water and a photoresistor placed on the opposite side. In a real-world implementation, a more precise turbidity sensor would be used. The ultrasonic distance sensor was used to detect whether a boat was anchored to the buoy. In the final system, this functionality would be replaced by a load cell to directly measure the force applied to the buoy, also avoiding any problems that may be caused by algae fouling on sensors.

The three LEDs were used as status indicators: green indicated that the buoy was available, orange indicated that it was reserved, and red indicated that it was occupied. The wind speed was measured using a small motor acting as a generator, while the temperature sensor was included to monitor the surrounding water temperature.

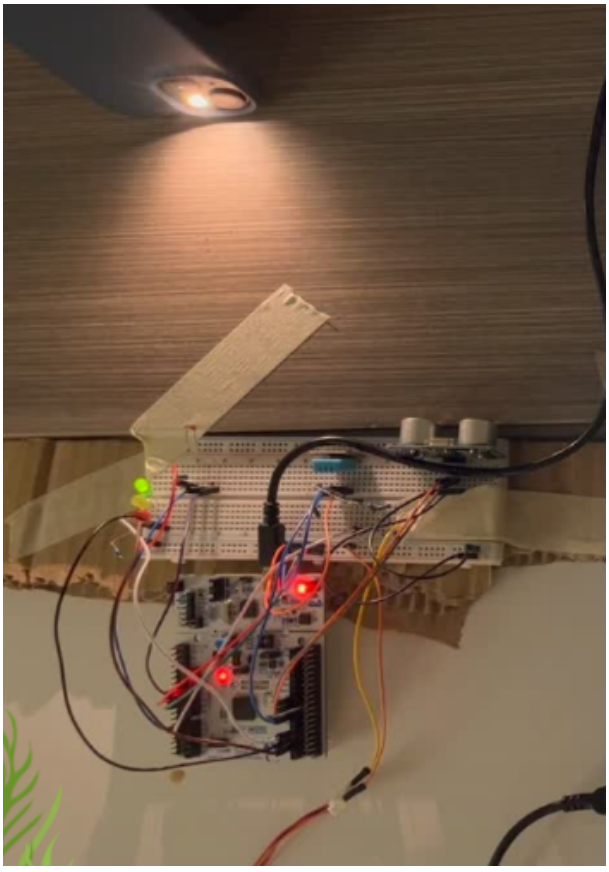


Figure 10: Buoy Prototype

The demonstrative video can be accessed through this link.

8 Conclusion

Throughout this project, we developed the concept of the PosiBuoy, a smart buoy system designed to balance the needs of boat owners and coastal ecosystem guardians while reducing the negative impact of anchoring on marine environments. By creating personas and user journeys, we identified the challenges and requirements of the different stakeholders and used these insights to define the key features of our solution.

The developed proof of concept demonstrated the feasibility of integrating sensors, communication, and status indicators into a buoy-based system. Although some components were simplified for testing purposes, the prototype successfully showed how environmental monitoring and buoy occupancy detection could be combined to support more sustainable anchoring practices.

Further development would focus on improving sensor accuracy, implementing a robust communication system, and testing the solution in real-world marine conditions. Overall, the PosiBuoy concept provides a promising approach to protecting coastal ecosystems while maintaining usability for recreational and professional boat users.

References

[1] EUobserver. *Mediterranean's hidden forests under siege by tourism*. URL: https://euobserver.com/40270/mediterraneans-hidden-forests-under-siege-by-tourism/ (cit. on pp. II, 1).

[2] France24. *Posidonia meadows: lungs of the seas at risk*. URL: https://www.france24.com/en/tv-shows/down-to-earth/20230824-posidonia-meadows-lungs-of-the-seas-at-risk (cit. on pp. II, 1).

[3] PileBuck. *Helical anchors for mooring applications*. URL: https://pilebuck.com/helical-anchors-superior-choice-mooring-applications/ (cit. on pp. II, 10).

[4] Sylvain Pioch and Aurore Léocadie. *OVERVIEW OF ECO-MOORING FACILITIES*. Tech. rep. International Coral Reef Initiative (ICRI), Jan. 2018. DOI: 10.13140/RG.2.2.27889.95846 (cit. on p. II).

[5] RAC/SPA. *Posidonia oceanica meadows documentation*. URL: https://www.rac-spa.org/sites/default/files/doc_vegetation/ramoge_fr.pdf (cit. on p. 1).

[6] WWF Mediterranean Marine Initiative. *Saving Mediterranean Posidonia*. URL: https://www.wwfmmi.org/what_we_do/blue_economy/protect_our_blue_forests/saving_mediterranean_posidonia2/ (cit. on p. 1).

[7] Matt McGrath (BBC). *Climate change: 'Forever plant' seagrass faces uncertain future*. URL: https://www.bbc.com/news/science-environment-56378397 (cit. on p. 1).

[8] Parc National des Calanques. *Promethee Med project*. URL: https://www.calanques-parcnational.fr/fr/actualites/lancement-du-projet-promethee-med-pour-la-protection-des-herbiers-de-posidonie (cit. on p. 1).

[9] Italian Ministry for the Environment and Energy Security (MASE). *Sea Forest Life project*. URL: https://www.mase.gov.it/portale/progetto-sea-forest-life-d-r-e-am-italia-soc-coop-agr- (cit. on p. 1).

[10] MSM Ocean. *Oceanographic buoys*. URL: https://msmocean.com/en/oceanographic-buoys/ (cit. on p. 7).

[11] GLOS. *Observing buoys*. URL: https://glos.org/observing/buoys/ (cit. on p. 7).

[12] Sensorex. *Dissolved oxygen sensors*. URL: https://sensorex.com/dissolved-oxygen/ (cit. on p. 8).

[13] LG Sonic. *Aquawiper: Biofouling prevention system*. URL: https://www.lgsonic.com/insights-aquawiper-no-biofouling-no-downtime-and-no-manual-cleaning/ (cit. on p. 8).

[14] UK Centre for Environment Fisheries and Acquaculture Science. *Extensively underwater camera network to monitor and protect ocean wildlife and blue economies launched across ten uk overseas territories*. URL: https://www.cefas.co.uk/news-and-resources/news/extensive-underwater-camera-network-to-monitor-and-protect-ocean-wildlife-and-blue-economies-launched-across-ten-uk-overseas-territories/ (cit. on p. 8).

[15] CEFAS. *SmartBuoy monitoring network*. URL: https://smartbuoy.cefas.co.uk/details/TH1/INT (cit. on p. 8).

[16] CatchCam. *Seagrass and blue carbon monitoring*. URL: https://catchcam.tech/blog/seagrass-blue/ (cit. on p. 8).

[17] LG Sonic. *FAQ About LG Sonic wiper system*. URL: https://www.lgsonic.com/faq/ (cit. on p. 8).

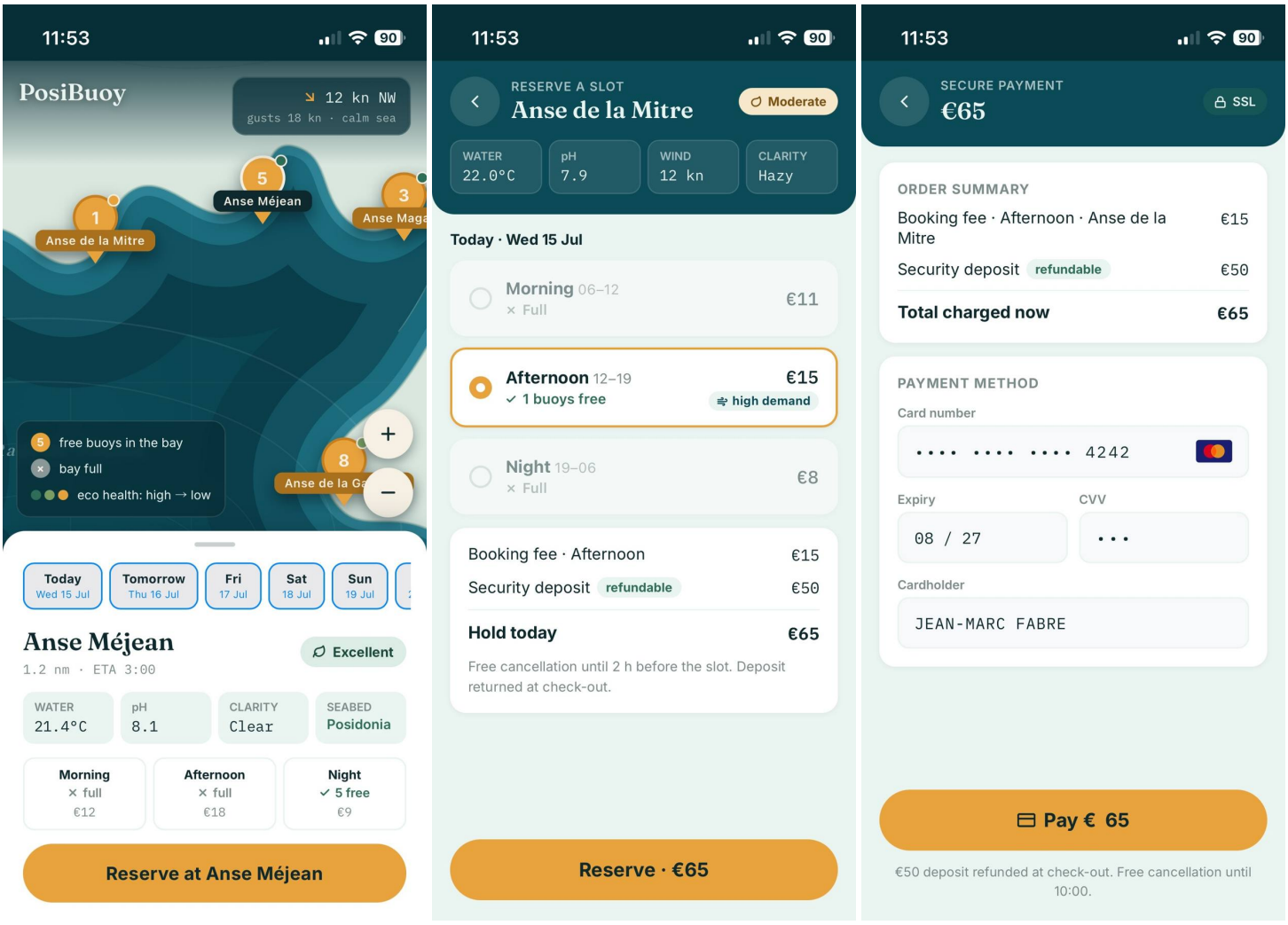
[18] Métropole Toulon Provence Méditerranée. *Zones de Mouillages et d'Équipements Léger*. URL: https://www.provencemed.com/en/activities/mooring-area-in-port-cros-5586657/ (cit. on p. 8).

[19] Kyocera and Nitto Seimo. *Smart sensing research buoy powered by ocean currents*. URL: https://www.industry-asia-pacific.com/news/92557-kyocera-and-nitto-seimo-

launch-trial-of-smart-sensing-research-buoy-powered-by-ocean-currents (cit. on p. 9).
[20] EcoAct. *Preservation of Mediterranean seagrass meadows*. URL: https://eco-act.com/fr/blog/preservation-des-herbiers-mediterraneens/ (cit. on p. 10).

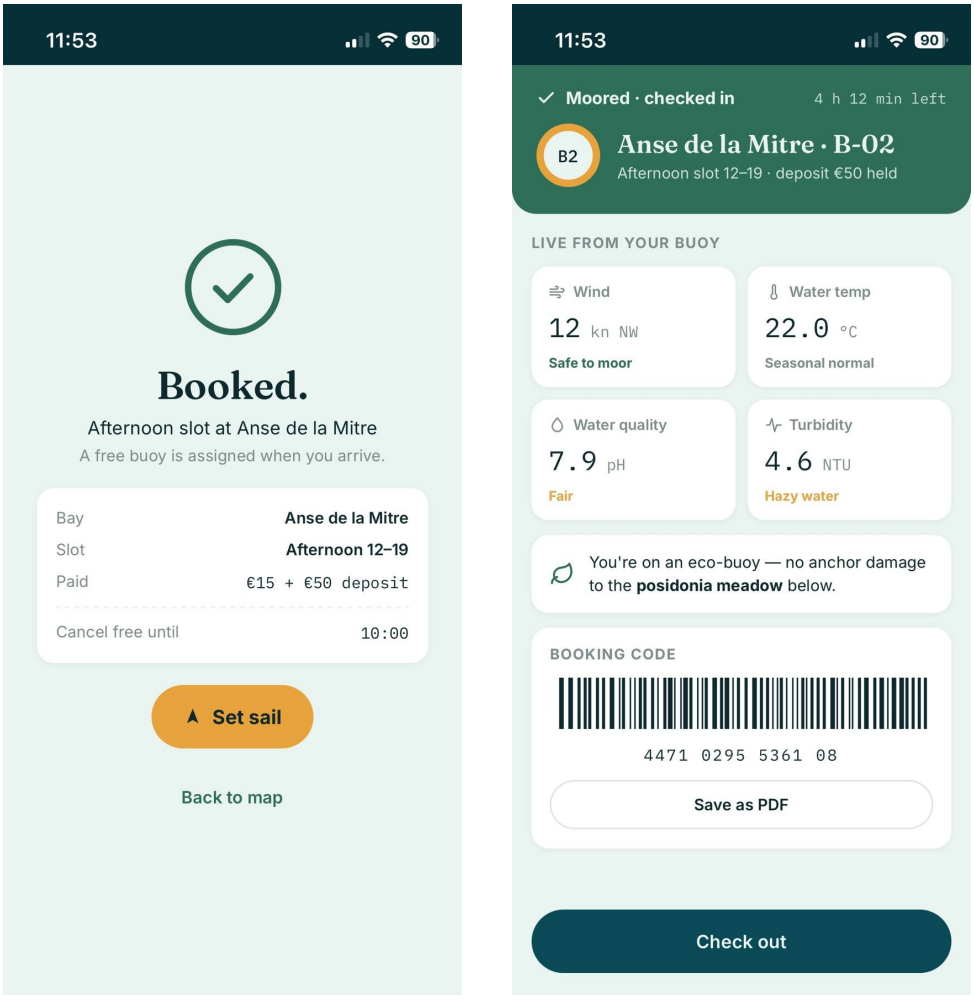
Appendix A: Mobile App Screenshots

The following screenshots present a mockup of the proposed mobile application's user interface. This prototype was developed to demonstrate the reservation workflow and intended user experience for boat owners using the PosiBuoy system.



(a) Home (b) Reserving (c) Payment

Figure 11: Mobile App Screenshots (Part 1)



(a) Booked (b) Confirmation

Figure 12: Mobile App Screenshots (Part 2)

Appendix B: Guardian Dashboard Screenshots

The following screenshots present the Guardian Dashboard interface. This prototype was developed to demonstrate the monitoring and management capabilities provided to marine park authorities.

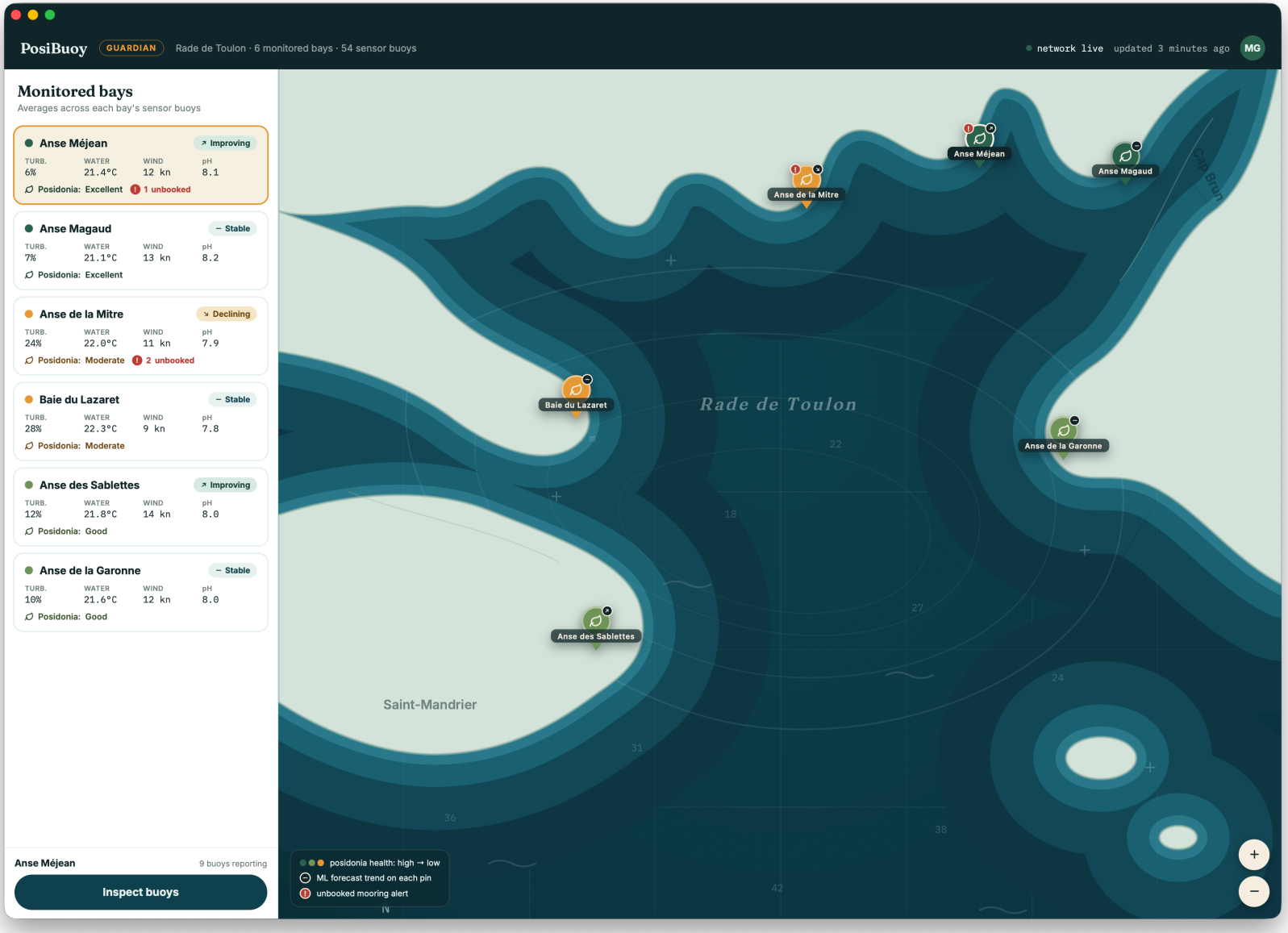


Figure 13: Guardian Dashboard: Home

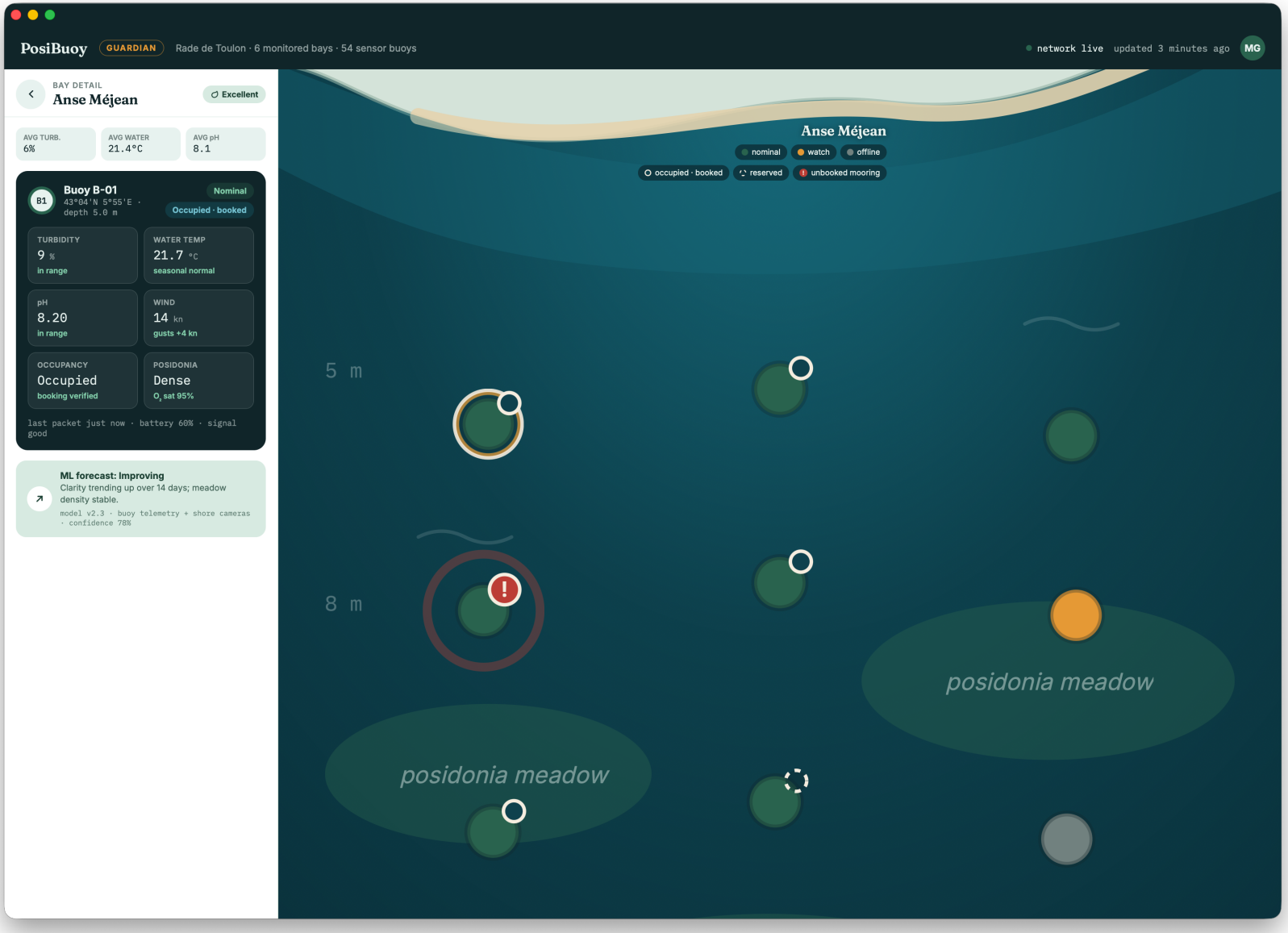


Figure 14: Guardian Dashboard: Bay Detail

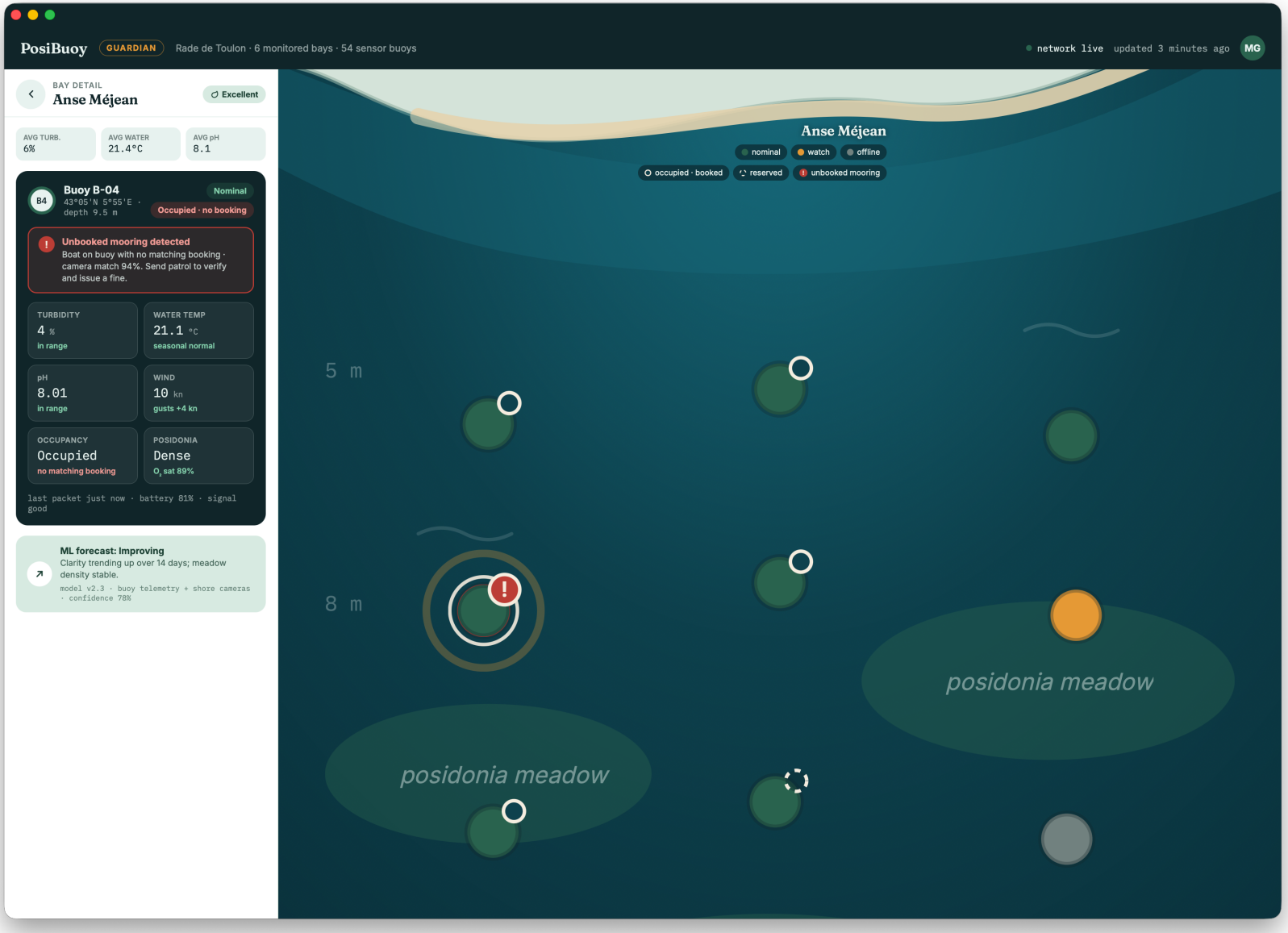


Figure 15: Guardian Dashboard: Unbooked Mooring

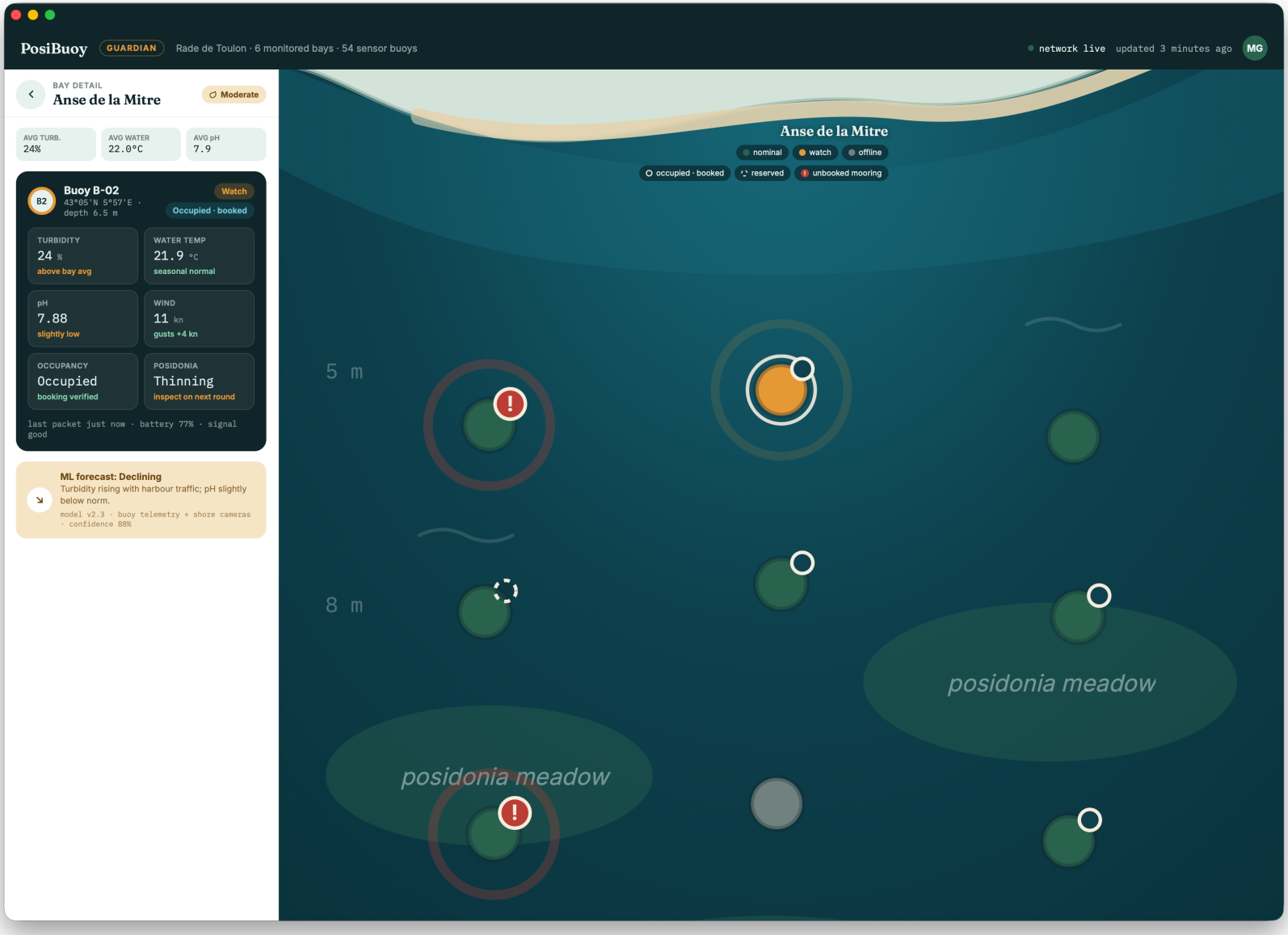


Figure 16: Guardian Dashboard: Declining Posidonia Alert